Lab report - Microscopy Imaging - How to build a virtual microscope

Group

November 9, 2026

1 Defining the Object

Q1.1 What would the image of such an object look like on a traditional widefield microscope? What is the minimum numerical aperture that should allow you to resolve the nuclei?

The image object is on an actial widefild microscope, a bunch of cells that have a certain thikness and so absorb a certain amount of light. The image is then genuinely a absorbtion spectrum, depending of the phase. If $\Delta z = 0$ (focus): A standard camera or the human eye detects only optical intensity $I = |E|^2 = 1$. The phase perturbations do not generate amplitude contrast when the object plane is conjugate to the detector.

Out of contrast : $\Delta z \neq 0$ Cell boundaries and the refractive index gradient between the nucleus and the cytoplasm become visible as bright and dark diffraction fringes (haloes) around the edges.

For the numerical apperture:

$$R_{\text{nuc}} = 50\,\mu\text{m} \times 0.10 = 5\,\mu\text{m}$$

$$\Leftrightarrow$$

$$D_{\text{nuc}} = 2 \times R_{\text{nuc}} = 10\,\mu\text{m}$$

So, we need to compute the diffraction limits. To resolve the object of diameter D_{nuc}, we must use two differents metrics:

- Abbe Resolution Limit:

$$d_{\text{min}} = \frac{\lambda}{2 \cdot NA} \leq D_{\text{nuc}} \implies NA \geq \frac{\lambda}{2D_{\text{nuc}}}$$

$$\Leftrightarrow$$
$$NA \geq \frac{0.55\,\mu\text{m}}{2 \times 10\,\mu\text{m}} = \mathbf{0.0275}$$

- Rayleigh Criterion:

$$r_{\text{Rayleigh}} = 0.61\frac{\lambda}{NA} \leq D_{\text{nuc}} \implies NA \geq 0.61\frac{\lambda}{D_{\text{nuc}}}$$
$$\Leftrightarrow$$
$$NA \geq 0.61 \times \frac{0.55\,\mu\text{m}}{10\,\mu\text{m}} \approx \mathbf{0.034}$$

So, the absolute theoretical lower bound to resolve the nucleus as an individual feature is $NA \approx 0.034$.

2 Imaging Contrast Mechanisms

Q.4 In brightfield, the unscattered light dominates and the transparent cells are barely visible (only the absorbing nuclei and a faint edge). In darkfield, the annular pupil blocks the unscattered (zero-frequency) light: the background becomes black and only light scattered by the cells reaches the camera, so the cells appear bright, mostly at their edges where the phase gradient (light deflection) is strongest. Darkfield reveals the cell outlines but does not give a quantitative phase measurement.

Q5 *Phase contrast (weak object transfer function).*

For a weak phase object ($\varphi \ll 1$), the field can be linearized:

$$E = e^{i\varphi} \approx 1 + i\varphi.$$

The unscattered component (1) and the scattered component ($i\varphi$) are in quadrature, so $|1+i\varphi|^2 = 1+\varphi^2 \approx 1$: the cells are invisible. A Zernike phase plate multiplies the zero frequency by i (quarter-wave shift), giving

$$E' = i + i\varphi = i(1+\varphi) \quad \Rightarrow \quad I = (1+\varphi)^2 \approx 1 + 2\varphi,$$

so the phase is converted linearly into intensity. Since our cells reach $\varphi \approx 30$ rad, far from the weak-object regime, the phase was rescaled to $\varphi_{\max} = 0.3$ rad to stay within the validity of the linearization. In brightfield, this weak object is only visible through its absorption (panel 3), whereas in phase contrast (panel 4) the cells appear as bright disks whose brightness follows the phase map.

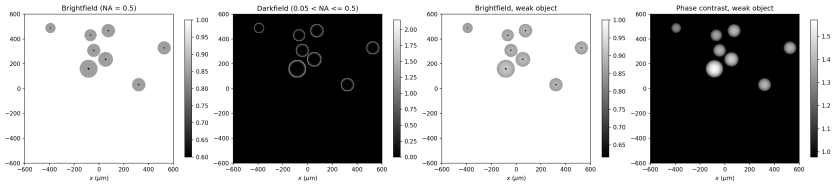


Figure 1: contrast images